

Internship Task Report: Task 4 Day 4

Date: [4.06.2026]

Intern Name: [Mahek.oza]

(Interns Batch):- Cyber security

Target Website: [https://kryosync.netlify.app](https://kryosync.netlify.app)

### **Task:- Top 10 cyber threats of 2026**

#### **The top 10 cyber threats of 2026 are as follows:-**

##### **1. Agentic AI & Super-Automated Hacking**

Attackers have shifted from using basic generative AI (like writing phishing emails) to deploying autonomous AI agents. These malicious agents can independently scan networks, discover vulnerabilities, write adaptive payloads on the fly, and execute multiple stages of a cyberattack without human intervention. This has drastically lowered the barrier to entry for low-skill threat actors to launch high-impact operations.

##### **2. Deepfake & Synthetic Identity Exploitation**

Deepfake technology has matured to a level where distinguishing real video and audio from synthetic media is incredibly difficult. Cybercriminals are using real-time voice cloning and video manipulation to impersonate executives in high-value Business Email Compromise (BEC) scams. Furthermore, state-sponsored actors are using sophisticated deepfake personas to successfully slip fraudulent remote workers onto Western payrolls for corporate espionage.

##### **3. Token Theft & Session Hijacking**

As multi-factor authentication (MFA) became standard, hackers adapted. Instead of targeting passwords, threat actors are weaponizing advanced info-stealers (like LummaC2) to harvest active session tokens. By stealing a token from a user's browser, attackers completely bypass MFA and log straight into cloud applications as an already-authenticated user.

##### **4. Over-Privileged SaaS & Cloud Supply Chain Cascades**

Modern organizations rely heavily on intersecting Software-as-a-Service (SaaS) and Cloud tools. Attackers target the “connective tissue”—the API integrations between these systems. If a single third-party vendor or tool with over-privileged API access is compromised, the blast radius can instantly cascade into hundreds of connected corporate environments.

## 5. Weaponization of Trusted Cloud Tooling

Adversaries are increasingly hiding in plain sight by using a tactic known as “living off the cloud.” Instead of using obvious malicious infrastructure, they use legitimate platforms like Google Calendar, Dropbox, and GitHub to host malware, camouflage command-and-control (C2) communications, and blend seamlessly into normal enterprise network traffic.

## 6. Ransomware 5.0 (Triple & Quadruple Extortion)

Ransomware has become highly targeted and automated. Beyond just encrypting files, attackers employ brutal multi-layered pressure tactics. They steal the data, threaten to leak it, launch massive Distributed Denial of Service (DDoS) attacks to take down the victim’s operations, and actively harass the victim’s clients or third-party associates until a ransom is paid.

## 7. Hyper-Volumetric DDoS Strikes

Fueled by next-generation IoT botnets, distributed denial-of-service (DDoS) attacks are breaking records. These hyper-volumetric strikes throw unprecedented amounts of junk traffic at servers, completely exhausting infrastructure capacity within seconds. This leaves zero time for manual human response, requiring entirely automated defensive mitigation.

## 8. Adversarial Attacks on Enterprise AI Systems

As companies rush to deploy internal AI models, retrieval mechanisms (RAG), and integrated workflows, those systems have become prime targets. Attackers are using

prompt injection, data poisoning (corrupting training data), and model manipulation to force company AI systems to leak sensitive backend data or execute unauthorized code.

## 9. Edge Device & IoT Vulnerabilities

With billions of connected devices across smart homes, corporate spaces, and smart cities, the attack surface is vast. Attackers routinely target unmanaged edge devices, smart infrastructure, and hybrid-work home routers. Because these devices often run outdated firmware, use default credentials, or lack proper endpoint security, they serve as easy initial entry points into larger networks.

## 10. Identity-Based API Probing

Because the cloud perimeter is defined by APIs, adversaries spend massive amounts of resources scanning for exposed, undocumented APIs or misconfigured Identity and Access Management (IAM) permissions. Once a weak point is found, they exploit it to quietly exfiltrate vast volumes of sensitive data without triggering traditional endpoint security alerts.

### **\*CIA triad:**

The CIA Triad is the foundational bedrock of cybersecurity and information security. It is a benchmark model designed to guide policies and security mechanisms for protecting data within an organization.

The acronym stands for Confidentiality, Integrity, and Availability.

### Core Pillars of the CIA Triad

#### 1. Confidentiality (Data Privacy)

Definition: Ensuring that sensitive information is only accessed by authorized individuals and kept hidden from everyone else.

Key Concept: Preventing unauthorized disclosure.

Common Controls:

Encryption: Converting data into unreadable ciphertext (e.g., AES-256) both at rest and in transit.

Access Control Lists (ACLs) & RBAC: Role-Based Access Control ensures users only see data required for their specific job.

Multi-Factor Authentication (MFA): Verifying identity via multiple independent credentials.

Real-world Threat: Data breaches, phishing attacks, or shoulder-surfing.

## 2. Integrity (Data Accuracy)

Definition: Ensuring that data remains accurate, complete, and unaltered throughout its entire lifecycle.

Key Concept: Preventing unauthorized modification or deletion.

Common Controls:

Hashing: Using algorithms like SHA-256 to create a unique digital fingerprint of a file. If even a single character changes, the hash changes completely.

Digital Signatures: Verifying the authenticity and origin of a file or message.

Version Control & Backups: Allowing systems to revert to a known valid state if data is corrupted.

Real-world Threat: Man-in-the-Middle (MitM) attacks altering a wire transfer amount, or accidental database corruption.

## 3. Availability (Data Accessibility)

Definition: Ensuring that authorized users have reliable, timely access to data and resources when they need them.

Key Concept: Preventing downtime and disruptions.

Common Controls:

Redundancy & Failover: Having backup servers, power supplies, and network paths ready to take over instantly if a primary component fails.

DDoS Protection: Scrubbing traffic to block malicious surges aimed at crashing web servers.

Regular Patching: Keeping software updated to prevent system crashes caused by bugs or exploits.

Real-world Threat: Distributed Denial of Service (DDoS) attacks, ransomware locking down systems, or physical power outages.